Battery Monitor

Release 1.0

TekTutor

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This project documents a 12 V battery monitor.

CHAPTER

ONE

OVERVIEW

The battery monitor reads the pack voltage, converts it into a state of charge and warns the driver when the pack runs low.

1.1 Voltage to state of charge

Pack voltage	State of charge	Low warning
12700 mV	100 %	No
12250 mV	50 %	No
11999 mV	22 %	Yes
11800 mV	0 %	Yes

Sphinx is published at https://pypi.org/project/Sphinx/ and Sphinx-Needs at https://pypi.org/project/sphinx-needs/.

CHAPTER

TWO

REQUIREMENTS

Requirement: **Report state of charge** *REQ_SOC*

status: open
tags: soc, display

satisfied by: *SPEC_SOC_LINEAR*
verified by: *TC_SOC_CLAMP*, *TC_SOC_RANGE*

The system shall report the state of charge of the 12 V pack as a whole percentage from 0 to 100.

Requirement: **Warn on low battery** *REQ_LOW_WARN*

status: open
tags: warning, driver

satisfied by: *SPEC_LOW_THRESHOLD*
verified by: *TC_LOW_BELOW*, *TC_LOW_EDGE*

The system shall warn the driver when the pack voltage falls below the low threshold.

Requirement: **Log voltage samples** *REQ_LOG*

status: draft
tags: diagnostics

satisfied by: *SPEC_LOG_FORMAT*

The system shall store every voltage sample with a timestamp for workshop diagnostics.

CHAPTER

THREE

SPECIFICATIONS

Specification: **Linear voltage to SOC mapping** *SPEC_SOC_LINEAR*

status: accepted
tags: soc

satisfies: *REQ_SOC*

Map 11800 mV to 0 % and 12700 mV to 100 % on a straight line. Clamp every value outside that range to 0 % or 100 %.

Specification: **Low threshold of 12000 mV** *SPEC_LOW_THRESHOLD*

status: accepted
tags: warning

satisfies: *REQ_LOW_WARN*

Report the pack as low when the last sample is below 12000 mV. A sample of exactly 12000 mV is not low.

Specification: **Sample log format** *SPEC_LOG_FORMAT*

status: draft
tags: diagnostics

satisfies: *REQ_LOG*

Store each sample as `<ISO 8601 timestamp>;<voltage in mV>`.

CHAPTER

FOUR

TEST CASES

Test Case: **SOC at mid range** *TC_SOC_RANGE*

status: passed

verifies: *REQ_SOC*

Feed 12250 mV. Expect 50 %.

Test Case: **SOC clamps outside range** *TC_SOC_CLAMP*

status: passed

verifies: *REQ_SOC*

Feed 13500 mV and expect 100 %. Feed 10000 mV and expect 0 %.

Test Case: **Low warning below threshold** *TC_LOW_BELOW*

status: passed

verifies: *REQ_LOW_WARN*

Feed 11999 mV. Expect the low warning.

Test Case: **No warning at threshold** *TC_LOW_EDGE*

status: open

verifies: *REQ_LOW_WARN*

Feed exactly 12000 mV. Expect no low warning.

CHAPTER

FIVE

TRACEABILITY

5.1 All needs

ID	Type	Title	Status	Tags
REQ_LOG	req	Log voltage samples	draft	diagnostics
REQ_LOW_W.	req	Warn on low battery	open	warning; driver
REQ_SOC	req	Report state of charge	open	soc; display
SPEC_LOG_F	spec	Sample log format	draft	diagnostics
SPEC_LOW_T	spec	Low threshold of 12000 mV	accepted	warning
SPEC_SOC_L	spec	Linear voltage to SOC mapping	accepted	soc
TC_LOW_BEL	test	Low warning below threshold	passed	
TC_LOW_EDG	test	No warning at threshold	open	
TC_SOC_CLA	test	SOC clamps outside range	passed	
TC_SOC_RAN	test	SOC at mid range	passed	

5.2 Needs tagged warning

ID	Type	Title
REQ_LOW_WARN	req	Warn on low battery
SPEC_LOW_THRES	spec	Low threshold of 12000 mV

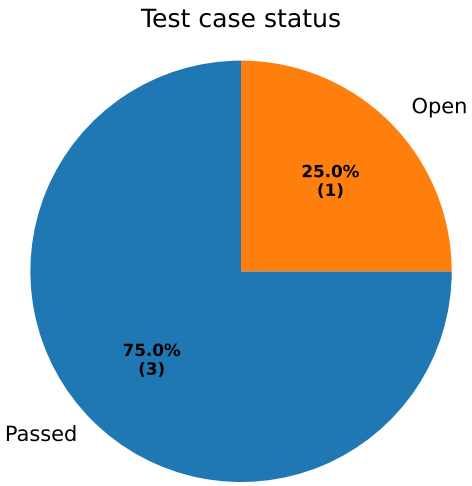
5.3 Open or draft needs

ID	Type	Title	Status
REQ_LOG	req	Log voltage samples	draft
REQ_LOW_WAR	req	Warn on low battery	open
REQ_SOC	req	Report state of charge	open
SPEC_LOG_FOI	spec	Sample log format	draft

5.4 Traceability graph

Fig. 1: Battery monitor traceability

5.5 Test status



5.6 Requirement list

REQ_SOC: Report state of charge (open) [soc; display]
REQ_LOW_WARN: Warn on low battery (open) [warning; driver]
REQ_LOG: Log voltage samples (draft) [diagnostics]